

Continuous Manufacturing of Recombinant Therapeutic Proteins: Upstream and Downstream Technologies



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Abstract Continuous biomanufacturing of recombinant therapeutic proteins offers several potential advantages over conventional batch processing, including reduced cost of goods, more flexible and responsive manufacturing facilities, and improved and consistent product quality. Although continuous approaches to various upstream and downstream unit operations have been considered and studied for decades, in recent years interest and application have accelerated. Researchers have achieved increasingly higher levels of process intensification, and have also begun to integrate different continuous unit operations into larger, holistically continuous processes. This review first discusses approaches for continuous cell culture, with a focus on perfusion-enabling cell separation technologies including gravitational, centrifugal, and acoustic settling, as well as filtration-based techniques. We follow with a review of various continuous downstream unit operations, covering categories such as clarification, chromatography, formulation, and viral inactivation and filtration. The review ends by summarizing case studies of integrated and continuous processing as reported in the literature.

Keywords Biomanufacturing, Continuous, Continuous chromatography, Downstream, Integrated, Perfusion, Upstream

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1 Introduction

Although the vast majority of current biotherapeutics are manufactured via upstream and downstream batch operations [1–3], continuous manufacturing is receiving serious consideration in the biopharmaceutical industry [4–9]. Over the past years, continuous technologies for individual unit operations have been developed and applied in both industry and academia. As these technologies have matured, examples of integration of continuous operations are emerging with increasing frequency. Ultimately, this progression could lead to fully integrated and continuous end-to-end bioprocesses. Such processes potentially offer several advantages, as enumerated below.

First, from a cost-of-goods perspective, continuous manufacturing unlocks process intensification and integration opportunities, increasing productivity and eliminating some unit operations (especially hold steps). Continuous operation also enables production at smaller scales, which in turn reduces equipment size and facility footprint, generates opportunities to apply disposable technologies, and ultimately leads to reduced capital and operational costs, as highlighted in a number of recent economic analyses [10–14]. If integrated and continuous systems can be reliably executed in a consistent fashion over time, they also become prime candidates for increased automation, leading to reduced labor costs [8].

Second, continuous manufacturing can also increase facility responsiveness and flexibility. The low residence times and cycle times of a well-designed continuous process should allow for the manufacture of both stable and labile products, and companies with such processes could hypothetically build standardized platforms that serve a wide variety of molecule types. Standardization in combination with small-volume, intensified processing makes possible nimble, multi-product ballroom facilities [15, 16].

Third, continuous processing offers several positive implications for product quality. As mentioned previously, lower residence times and reduced holds should generally lead to better and less variable product quality. Additionally, because the regulatory definition of batch primarily depends on product uniformity and not the mode of manufacturing (21 C.F.R. § 210.3, 2015), a continuous process developed to perform consistently and steadily (with respect to product quality) both within and across runs can be leveraged to provide enormous operational freedom because pooling/splitting/batching strategies can be decoupled from run day [15, 17].

This review relates the progress made toward continuous biomanufacturing of therapeutic recombinant proteins. We first discuss approaches for continuous cell culture, with a focus on perfusion-enabling cell separation technologies. We follow with a review of various continuous downstream unit operations, covering categories such as clarification, chromatography, formulation, and viral inactivation and filtration. Relevant technologies and experiences from other related industries are also discussed. The chapter ends by summarizing case studies of integrated and continuous processing in the literature.

2 Current Continuous Technologies

2.1 Continuous Cell Culture

A holistic upstream/downstream integrated and continuous process requires a cell culture platform with a continuous (or at least semi-continuous) output of product-containing harvest. Fed-batch bioreactors are the current industry standard for recombinant protein production [3] but, because of their inherent batch nature, they are not readily amenable to a continuous process. However, several alternative cell culture strategies have been developed over the past 20 years that are capable of continuous product output and integration with downstream operations [4, 5, 18–20].

Continuous cell culture operations can offer more than just a continuous source of product. In many situations their longer durations and/or higher cell densities allow for process intensification relative to fed-batch approaches [14]. In some instances, continuous systems can also be controlled consistently over long durations, simplifying operations and guaranteeing consistent product quality attributes [15, 17, 21].

2.1.1 No Cell Retention

The most straightforward approach to continuous cell culture is the chemostat, where fresh medium is fed into the bioreactor and spent medium containing both cells and product is removed at the same rate. No separation devices are used to retain cells, so the cell concentrations in the harvest stream and bioreactor are equal. Although chemostats for mammalian culture have been studied for more than half a century [22, 23], they are still most commonly applied as experimental tools at small scales [24–27]. There is at least one example of a chemostat-like commercial process: Factor IX is made in a suspension cell culture using a semi-continuous batch re-feed process [28]. Because of its lability, the product cannot reside in the bioreactor for long durations. The batch re-feed process allows product quality to be controlled and maintained and reactor turn-around to be reduced.

However, the low growth rates of mammalian cells prevent such systems from achieving high cell densities and productivities, rendering chemostats infeasible and undesirable for manufacturing. To achieve productive continuous culture, some form of cell retention must be employed to intensify cell density in the bioreactor and decouple the cell growth rate from the media addition rate. Several categories of these perfusion-enabling cell retention devices, each with advantages and disadvantages, are discussed below.

2.1.2 Immobilized Cell Retention

Cells can be permanently separated from the perfusate if they can be immobilized to a fixed matrix. Culture medium can then be passed through the cellular environment and spent product-containing medium can be collected at the other end. Immobilized systems do not suffer from washout at any perfusion rate, cells are generally protected from shear, and high packed cell densities are possible. Common disadvantages of immobilization include cell-to-cell heterogeneity, representative sampling and accurate biomass monitoring, poor oxygen and nutrient transfer, and difficult scale-up [18]. Examples of immobilized systems include hollow-fiber membranes, flat plate membranes, gel encapsulation, ceramic matrices, and fluidized beds. Several comprehensive reviews of these strategies can be found in the scientific literature [29–33].

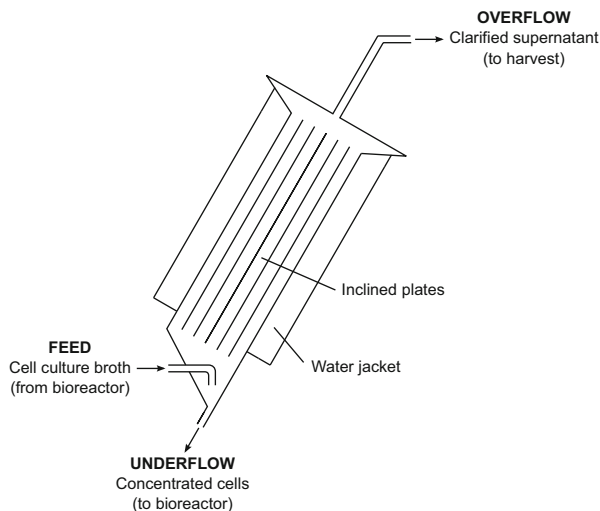
2.1.3 Density-Based Cell Retention

Viable, suspended mammalian cells have a density greater than that of their surrounding supernatant, and various techniques have been developed to exploit this difference to extract a relatively cell-free harvest, simultaneously retaining cells within a bioreactor. Devices that employ density-based separation include gravitational settlers, centrifuges, acoustic settlers, and hydrocyclones.

Gravitational Settling

Gravity can be exploited to separate cells from harvest if the culture can be pumped upward at a rate slower than the cell settling velocity. At standard conditions this approach only enables low perfusion rates although requiring long residence times for cells in the separator, away from the controlled conditions of the bioreactor. Early implementations of gravitational settlers included a conical sedimentation column [34, 35], an intra-reactor settling zone [36], and a Dortmund settler [37]. Although these systems could be maintained for long durations (25 days and beyond), they could not reach densities above 7 million cells/mL. Average viabilities were also fairly low (45–85%), presumably because of the long residence time of cells in the separator.

Fig. 1 Diagram of an inclined cell settler [43]



To increase cell settling velocity, enable higher perfusion rates, and decrease separator residence time, cells can be cultured on suspended microcarriers [38–40]. However, microcarrier-based cultures present a number of difficulties, including heterogeneity within the bioreactor because of stratification, cell-to-cell heterogeneity because of layered growth on microcarriers, and increased sensitivity to shear from agitation and sparging [41, 42].

Alternatively, settler geometry can be optimized. As described in Fig. 1, inclined settlers with sloped, parallel plates have been designed to create laminar flow and leverage the Boycott effect to allow cells to settle, accumulate, and descend down the lower plate with enhanced separation efficiency [44, 45]. Residence time can be further decreased by cooling and vibrating the settler, as well as flushing periodically with gas [46]. Mathematical and computational modeling can be used to explore various geometries and optimize inclined settler design [47, 48]. Methodology for empirically determining cell settling velocity has also been developed, further informing design decisions [49].

Gravitational settlers are well-suited to long-duration operation as there is no physical barrier prone to fouling or clogging. There are several examples of run lengths over 40 days [50, 51] and commercial examples running up to 6 months [21, 52]. Another commonly cited advantage of gravity settlers is their ability to retain viable cells and remove smaller dead cells via the perfusate [53].

Gravitational settlers have also been demonstrated to scale up relatively well in manufacturing settings. Inclined settlers have recently been used to develop high-density N-1 seed train operations to increase the inoculation density of the production bioreactor to reduce durations and increase facility utilization. After testing at laboratory scales, an inclined settler was implemented alongside a 3,000-L bioreactor and operated at a maximum perfusion rate of one reactor volume per day [54, 55].

Perfusion by gravitational settling does have disadvantages. Cell densities have only been demonstrated up to 20–30 million cells/mL; above that range, retention efficiency dramatically drops and cells pass through to the harvest. Separator residence times, even after optimization, are still fairly long [46]. Finally, because debris and some cells do exit the bioreactor in the harvest stream of a gravitational settler, an additional clarification step is required if downstream purification operations are integrated.

Centrifugation

Similar to gravitational settling, centrifugation relies on density differences for cell retention, but with an increased driving force that hypothetically should result in reduced residence times outside of the controlled bioreactor. Early experiments demonstrated both intermittent and continuous centrifugation using an autoclavable rotor connected to the reactor system via mechanical seals [56–58]. These cultures typically averaged 5–10 million cells/mL, peaking at around 15 million cells/mL, with durations up to 40 days. Perfusion experiments have also been conducted with disk stack centrifuges, where cells are separated from supernatant using a disk stack, collected in peripheral pouches, and sent back through the center of the rotor to the bioreactor [59, 60].

To eliminate the need for rotor sterilization and reliance on mechanical seals, the Centritech centrifuge (commercialized today by PneumaticScaleAngelus) was introduced. The Centritech uses continuous orthogonal flow and features a disposable insert (fixed onto a conical rotor) for cell culture collection and separation. The insert is connected to the bioreactor via a seal-free tubing bundle with an anti-twist mechanism allowing it to rotate at the centrifuge and remain stationary at the other end. Similar to gravitational settlers, the Centritech has been shown to retain viable cells preferentially and remove dead cells. Initial studies, however, did not lead to viable cell densities above 10 million cells/mL, and, at these higher densities, viabilities were quite low, averaging around 50% [61, 62]. More recent experiments have overcome these early challenges, resulting in high viabilities [54, 63]. One implementation in particular demonstrated average cell densities of 30 million cells/mL for over 30 days [64]. The Centritech has also been implemented commercially for the production of labile proteins [5].

Centrifuges that utilize continuous counterflow are also commercially available, including the Elutra cell separation system (Terumo BCT) and kSep (KPI Biopharma) [65]. Experience with these systems integrated to perfusion bioreactors is limited, but the kSep system has been demonstrated for up to 6 days at a peak of approximately 15 million cells/mL [66].

Acoustic Settling

Acoustic settlers trap cells in the pressure nodes of standing ultrasonic waves. Once trapped, cells aggregate and then settle back into the reactor by gravity [67]. Acoustic settlers do not require moving parts, can be easily cleaned and sterilized, and do not require any physical barrier for cell separation and therefore are not prone to fouling [68].

Different designs have been tested, including a dual-chamber device for integrated cooling [69, 70] and BioSep, a single-chamber approach marketed by Applikon [71, 72]. Periodic air backflush has also been incorporated to reduce pump stress and decrease cell residence time in the separator [73]. Investigators have explored the effect of various system parameters on retention efficiency and culture success, including power input, cell concentration, perfusion rate, bleed rate, cooling air flow, backflush frequency, backflush volume, duty cycle stop time, and recirculation ratio [74–77].

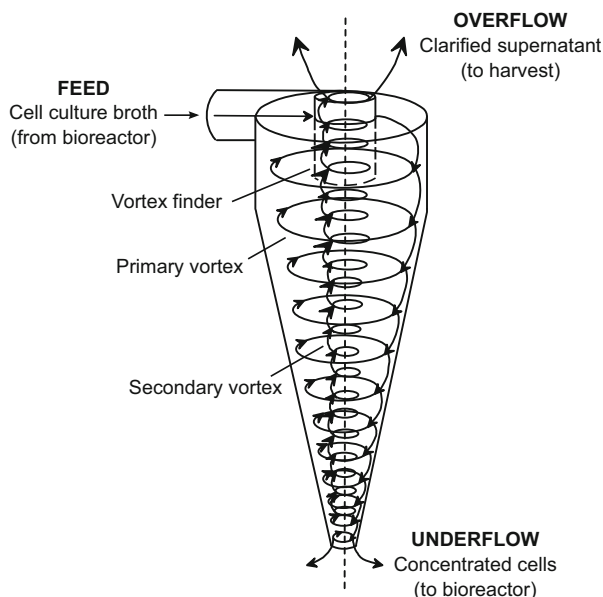
Acoustic separation has been successfully demonstrated at the pilot scale, enabling perfusion rates up to 200 L/day for 100-L bioreactor working volumes [76, 78]. Larger manufacturing devices are available and capable of handling 2,000 L/day, but no experience is reported in the literature. Similar to gravitational settlers, acoustic settlers are capable of long durations – experiments have persisted for 50 and even 100 days [73, 79]. However, densities have typically been demonstrated only at 10–15 million cells/mL, and the few examples that have transcended 20 million cells/mL [74, 80, 81] have also had very low viabilities between 40–60%, when reported.

Hydrocyclone Cell Retention

A hydrocyclone is a simple conical device that uses centrifugal forces to separate components by density. Figure 2 depicts this separator in a perfusion application: cell culture enters at high speeds and circles the cone. Cells are preferentially drawn outward and downward to the bottom outlet via a primary vortex as excess supernatant at low cell concentrations exits the top outlet via a secondary vortex [83]. Hydrocyclones are low-cost, have no moving parts, and can be readily applied to both disposable and clean-in-place/steam-in-place systems. These devices are also extremely size-efficient: a small hydrocyclone (1 cm diameter) can theoretically process up to 1,000 L/day [4]. Smaller volumes require intermittent perfusion because further scale-down of the hydrocyclone is not feasible [84].

Numerical models have predicted that hydrocyclones are capable of 90% cell retention efficiencies [82]. Short-duration experiments (up to 2 days) confirmed this estimate, showing 97% separation efficiency with no measurable impact on cell viability or productivity [85, 86]. More recently, longer cultures up to 8 days have been demonstrated, at bioreactor scales up to 300 L [87, 88]. Although hydrocyclone-mediated perfusion is a promising area, cell densities have yet to be pushed beyond 10 million cells/mL; additional study and work is required for

Fig. 2 Diagram of a hydrocyclone for cell retention [82]. The vortex finder is a short, removable section that prevents short-circuiting of feed directly into the overflow



this technology to compete with more conventional approaches such as gravitational settling or filtration.

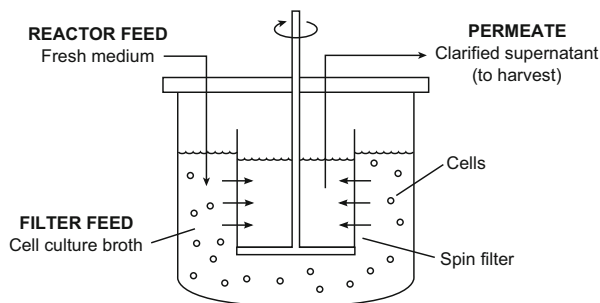
2.1.4 Size-Based Cell Retention

Various forms of filtration have been tested in perfusion systems. By using a filter as a perfusion-enabling cell separator, two unit operations (cell retention and clarification) can be integrated to deliver significant process intensification. Filtration-based perfusion also significantly eases downstream integration: because the harvest stream is clarified, and continuous purification operations (such as continuous chromatography) can be directly connected if flow rates are matched. The primary challenge is filter fouling: at medium-to-high cell densities, dead cells and debris accumulate and standard filtration approaches lack longevity. A number of different filtration technologies have been developed in an effort to meet this challenge, and are described below.

Spin Filtration

Spin filtration (Fig. 3) was an early solution to the problem of filter fouling where cell culture supernatant is pumped into a cylindrical filter and out of the bioreactor. The spin filter rotates rapidly (either internal or external to the reactor), preventing

Fig. 3 Diagram of a spin filter within a bioreactor



cells from fouling the membrane. When internal, the filter is often integrated onto the agitator shaft [89–91].

Although spin filters certainly offer superior performance relative to static dead-end filtration, filter fouling is still a significant concern. Fouling is worsened by charged filter surfaces and serum-based media [92], antifoam addition [93], high perfusion rates, high cell concentrations, low filter rotational velocities, and low filter surface area [94]. Models have been developed (both numerically and using computational fluid dynamics) to indicate the existence of substantial exchange rates across the filter surface that can contribute to fouling [95, 96].

An early study avoided fouling concerns by coupling a large-pore (120 μm) spin filter with microcarrier-seeded cell aggregates (200–600 μm), enabling a culture duration of 30 days with cell densities as high as 60–70 million cells/mL [97]. Another experiment maintained 18 million cells/mL for 35 days, peaking around 25 million cells/mL [64]. For the wide majority of cases, however, spin filter cultures reach an average of 10–15 million cells/mL [54, 71, 93, 94, 98, 99].

Scale-up is also challenging, as the radius of the cylindrical screen needs to increase quadratically with the radius of the bioreactor to maintain a constant filtration flux. The technology has been scaled to enable perfusion at pilot-scale bioreactors with 175-L [100] and 500-L [94] working volumes. Additional scale-up may require moving the filter apparatus external to the reactor [98].

Hollow-Fiber Filtration

Hollow-fiber filters were introduced as cell retention devices that could alleviate some of the concerns around spin filters. Cylindrical cartridges packed with hollow-fiber filters scale much more readily because filter surface area increases quadratically with cartridge radius. Early implementations placed these cartridges directly within the bioreactor and cell densities as high as 90 million cells/mL were demonstrated [101, 102], though most examples are external to the reactor, allowing for easier access, monitoring, and replacement.

Hollow-fiber cartridges can be operated using tangential-flow filtration (TFF), where feed flow travels tangentially to the filter surface and permeate flows perpendicularly (Fig. 4A). This cross-flow action reduces the trapping of cell debris

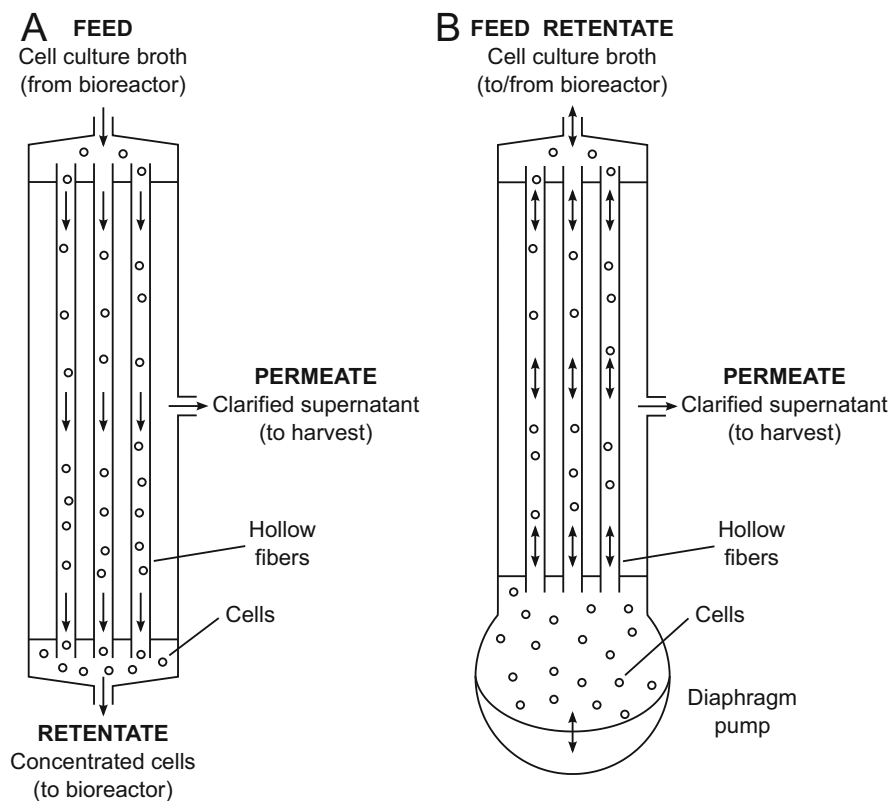


Fig. 4 Diagrams demonstrating cell retention via (A) tangential flow filtration and (B) alternating tangential flow filtration

and other solids into the filter, increasing system longevity. Early efforts established the feasibility of the technology across a variety of filter types and suppliers [103–105]. Filter fouling leading to product retention was noted across a membrane with 0.2 μm pore size [106]. One solution to increasing filter longevity was an increased recirculation rate and periodic pulsing through the hollow fibers [107]. Other researchers found that by avoiding high transmembrane pressures and washing the membrane surface daily at high flow rates with cell-free medium, fouling could be minimized and a continuous culture could be maintained for 30 days with a peak viable density above 30 million cells/mL [108].

More recently, cell densities and durations have been pushed increasingly higher and longer in TFF systems. Two different groups have demonstrated durations longer than 25 days peaking at 60–70 million cells/mL [109, 110] and Clincke achieved 45 days of operation with a peak density of 214 million cells/mL. At that ultra-high density, the viscosity of the culture and the dissolved pCO_2 became limiting [111].

Alternating tangential-flow (ATF) filtration is a specific implementation of TFF that has been widely applied to perfusion. A diaphragm pump is coupled to a hollow-fiber cartridge, and cell culture is alternately pumped into and out of the filter from the bioreactor (Fig. 4B). Advantages relative to standard TFF include a compact design and small footprint, gentler pumping, only one connection point necessary between the reactor and separation device, and significant back flow across the filter membrane which may reduce filter fouling. Despite these changes, filter fouling is still observed, and efforts have been taken to correlate and model filter performance [112].

High-density perfusion via ATF has been demonstrated for over 25 days [64, 109, 113], with one system operating for almost 70 days at greater than 40 million cells/mL [17] and another peaking at 130 million cells/mL [111]. ATF filtration has been frequently considered as a means toward high-density inoculation of the production bioreactor. Different groups have shown that inoculation densities of 5–10 million cells/mL are possible with no negative impacts on growth or product quality if the N-1 is operated with ATF perfusion [114–116].

Other Filtration

Disposable floating filters are available with some disposable bioreactor bag systems. These have been used to grow cells to densities above 20 million cells/mL, maintaining viabilities above 90% [117]. However, because of the lack of any cross flow, these filters are prone to clogging and are typically only used on the order of 10 days [118, 119].

Vortex flow filtration (VFF) consists of an internal, rotating cylinder surrounded by a concentric filter. Cell culture is sent to the annular space and cell-free supernatant is drawn through the filter and out of the annular space. The internal rotating cylinder forms Taylor vortices which prevent concentration polarization and particle deposition on the filter. Studies have been conducted both with hydrophilized polysulfone membranes [80] and a 10- μm stainless steel screen [120]. Low-density, long-duration runs have been demonstrated (5–10 million cells/mL for 75 days) as have shorter, higher density experiments (20 days with a 50 million cells/mL peak). As with spin filtration, scale-up could present challenges as the VFF diameter needs to increase quadratically with the bioreactor diameter.

2.1.5 Cell Density Control

Cell density can be controlled in perfusion systems to enable longer-duration cultures, increasing process consistency. If cells are left to proliferate unchecked, large swings in the extracellular environment can occur, potentially causing shifts to metabolism, productivity and product quality. The most common method for cell density control is to remove cell-containing fluid continuously or semi-continuously from the bioreactor at low rates [121]. This removal is commonly

referred to as “cell bleed” or “cell discard.” If the bleed rate can be controlled to equal the apparent growth rate (the difference between the true growth and death rates), then the cell density remains constant [74]. In filtration-based perfusion, where dead cells and debris accumulate in the bioreactor, cell bleed can also play an important role in removing these types of materials from the system.

One approach consists of setting a constant volumetric bleed rate and then allowing the culture to adjust itself until the apparent growth rate equals the bleed rate [78, 106]. The primary advantage of this method is its simplicity – no cell density measurements (online or offline) are required for implementation. The method’s downside is that, if cell growth rates change over time, the absolute cell densities fluctuate accordingly.

To incorporate feedback control, regular (e.g., daily) bolus bleeds can be manually executed after offline viable cell density measurements to bring the culture semi-continuously back to set point [110, 111]. Although this is relatively straightforward to implement from an infrastructure standpoint, the recurring requirement for manual intervention can be laborious. Additionally, because control is not continuous, by definition there are swings in cell density (and therefore cellular environment) between bleeds.

If a robust, online cell counting method can be developed, continuous bleeding for automated cell density control can be achieved. Various online proxies for cell density have been demonstrated, including optical density [122], metabolic rates [123–126], oxygen uptake rate [101], capacitance [17, 127, 128], and Raman spectroscopy [129, 130]. By linking the online measurement to a feedback-controlled bleed pump and recalibrating using offline measurements as necessary, cell densities can be maintained at consistent levels for long durations. Processes employing a feedback-controlled bleed need to be developed to optimize growth rate; if growth is too low, a consistent bleed cannot be maintained, and if it is too high, substantial product is sent to waste.

2.1.6 Historical Progress of Perfusion Technologies

Viable cell density profiles from various perfusion experiments with reported data in literature have been analyzed and compiled in figure form. In Fig. 5A, average viable cell densities in perfusion systems are plotted over time and categorized by separator type. Average viable cell density is defined as the integrated viable cell density (over the course of an experiment) divided by culture duration, and can serve as a rough indicator of a system’s potential for productivity. (Note that this metric also rewards cultures with longer durations, as the impact of the lower-density growth phase is minimized.) The time profile of average densities reveals that, with only limited exceptions, average perfusion densities did not exceed 20 million cells/mL until 2010, after which a large number of demonstrations at or above 30 million cells/mL have occurred. Cell retention via filtration is the predominant enabler of these higher-density systems.

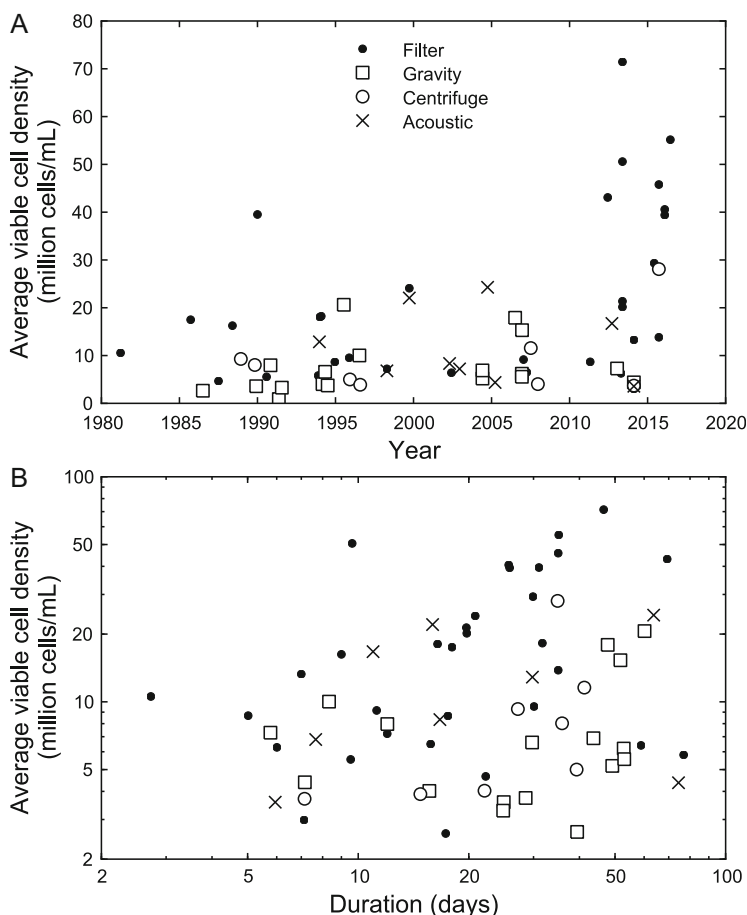


Fig. 5 Average viable cell densities reported in literature plotted by (A) year and (B) culture duration. Cell separator type is indicated by symbol (filtration – closed squares, gravitational settling – open squares, centrifugal settling – open circles, and acoustic settling – crosses). Data is based on over 60 experiments reported in the literature since 1980 [50, 53, 71, 74, 78, 88, 94, 97, 103, 111, 37, 54, 56, 73, 76, 93, 107, 122, 34, 62–64, 108, 109, 121, 131, 132, 51, 55, 80, 101, 110, 114, 124, 126, 133, 134, 17, 35, 36, 46, 57, 58, 81, 90, 91, 99, 102, 105, 113, 116, 117, 119]

In Fig. 5B, average viable cell density is plotted vs run duration. This figure reveals that most long-duration experiments (e.g., longer than 40 days) have occurred via density-based retention methods (gravitational, acoustic, and centrifugal settling). Because these methods do not utilize a physical barrier, they are not prone to fouling, and can more easily reach long culture times. However, when comparing systems by average density, it is clear that filtration-based perfusion has the advantage. At least within this curated data set, 75% of experiments with average densities over 20 million cells/mL and 100% over 30 million cells/mL used some form of filtration for cell retention (spin filtration, TFF, ATF, etc.).

2.2 *Continuous Chromatography*

Despite increasing competition from non-chromatographic techniques and pressure to reduce costs, chromatography has been and continues to be a dominant technique in biopharmaceutical purification. This prevalence is largely because of the high-resolution purification that can be achieved using chromatography, even for similar components. However, with improved cell lines and increasing titers in recent years, traditional batch chromatography has had some major drawbacks related to large-scale biopharmaceutical purification [135]. These include (1) the often inefficient usage of chromatography resin, (2) the large volumes of buffers needed, (3) the limited purity obtained for components with small adsorption differences, and (4) the discontinuity of the process. To overcome these limitations of batch chromatography, a number of continuous chromatography methods have been developed. Continuous chromatography operation can, in principle, be achieved by two modes of contacting the mobile and stationary phase: crosscurrent and countercurrent schemes. In certain cases, continuous operation can also be performed through use of flow-through chromatography with redundant units.

Continuous chromatography is particularly applicable to two main areas of protein purification: capture operations, for separating the product from cell culture harvest fluid, and polishing steps, where resolution between impurities and the protein of interest is important. For capture applications, integration of continuous chromatography systems with clarified harvest streams from a perfusion or fed-batch bioreactor can be envisioned. To enable continuous processing, the chromatography system must be capable of receiving a continuous feed stream. The output (eluate) can be continuous (such as with annular or flow-through chromatography) or periodic/cyclic (such as with periodic countercurrent chromatography). However, integration for prolonged periods of time (e.g., with a perfusion bioreactor) could be challenging, as it requires the continuous chromatography systems to be amenable to closed operation [15, 16, 136]. Some of the technologies that allow continuous chromatography are presented below.

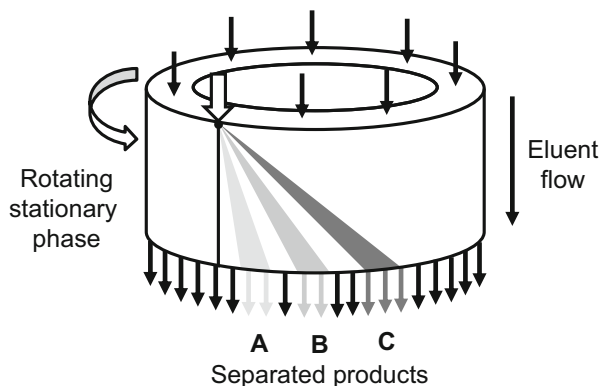
2.2.1 *Continuous Crosscurrent Chromatography*

In crosscurrent chromatography, the resin bed moves perpendicular to the direction of the liquid flow within the bed. In continuous crosscurrent systems, a steady-state separation occurs in the axial and circumferential direction, in contrast to conventional batch systems where separation occurs only in the axial direction [137].

Continuous Annular Chromatography

Continuous annular chromatography (CAC) allows truly continuous separation of multicomponent mixtures using a rotating annular bed composed of a

Fig. 6 Schematic view of P-CAC system showing isocratic elution of products A, B and C



chromatography matrix. This system was first conceived and proposed by Martin [138] and the theoretical feasibility was demonstrated by Giddings [139]. The first true CAC apparatus was constructed by Fox et al. [140, 141] and was tested for the purification of cow heart myoglobin and the separation of skim milk proteins from lactose and salt [142]. Since then, this technology has been applied to purify several biological molecules including sugars, amino acids, plasma-derived and recombinant blood clotting factors, plasmid DNA, antibodies, and vaccines.

The heart of the CAC system is a bed of conventional adsorbent which is packed between two concentric cylinders that forms the annulus (Fig. 6). The annular bed slowly rotates about its vertical axis. The feed mixture to be separated is introduced continuously through a stationary nozzle on top of the moving adsorbent bed. In the simplest case of isocratic chromatography, the rotating annulus is flooded with elution buffer which is flowing in the direction of feed (crosscurrent flow). Different components of the feed separate in the axial direction and develop helical bands from the fixed inlet point to the bottom of the annular bed. Several fixed outlets are kept at the bottom of the bed where the separated components are continuously recovered. The angular displacement of each component band from the feed point (i.e., the slope of the helical band) is dependent upon the flow rate, rotational speed of the annulus, and the component affinity for the stationary phase.

Step or gradient elution has also been applied for purification using CAC [136, 143–145]. In this case, the feed mixture and all necessary buffers (wash, elution, regeneration, etc.) are introduced through stationary nozzles along the circumference of the annulus. As the annulus rotates, each section of the stationary phase goes through load, wash, elution, and regeneration steps. Similar to isocratic chromatography, product can be continuously recovered from fixed outlet points.

In the late 1990s, CAC instruments became commercially available under the name P-CAC from Prior Separation Technology (Götzis, Austria). Several authors have successfully demonstrated transformation of conventional batch chromatography methods to CAC instruments [136, 144, 146]. Preparative continuous annular chromatography has been used for several different separation media, including ion exchange [143, 145, 147–149], displacement [150], affinity [144], and size

exclusion chromatography (SEC) [151–153]. The combination of two different separation media (SEC and ion-exchange chromatography) in one CAC column has also been demonstrated [154]. The potential for use of CAC for continuous integrated capture directly from perfusion cell culture was described by Vogel et al. [136] using an autoclavable P-CAC prototype. Despite several advances with CAC, this technology is yet to be implemented for large-scale protein purification. Obtaining uniform column packing and flow distribution and long term durability of the rotating mechanical parts are some of the remaining challenges with this technology.

Continuous Radial Flow Chromatography

Another adaptation of continuous annular chromatography is continuous radial flow chromatography (CRFC). CRFC operates similarly to CAC systems except feed and all necessary buffers flow in the radial direction (inward from periphery or outward from center in horizontal direction) instead of the axial direction (along the axis). The effective bed height of a radial flow column is the distance between the inner and outer cylinders and the cross sectional area is associated with the surface area of the tube. Short bed depth and large column cross-sectional area enable separations at high flow rates while maintaining low pressure drops [155]. However, radial flow columns have limited resolution because of the shorter bed depth. Thus, they are not ideally suited for applications where high resolution separation (e.g., analytical HPLC columns) or longer bed depth are required (e.g., SEC) [156].

Radial flow chromatography in batch mode has been investigated for several different separation media [157–160]. Both experimental results and theoretical modeling indicate that a radial flow column behaves similar to an axial flow column of shorter bed height (with similar resin volume) [157, 161–163]. Radial flow chromatography systems for batch operations are marketed by Sepragen (Hayward, CA) and Proxcys (Nieuw-Amsterdam, Netherlands). Recently, a Proxcys radial flow column packed with larger chromatography resin beads was utilized for capture of His-tag recombinant proteins from unclarified mammalian cell feeds [158]. Lay et al. [164] constructed a prototype CRFC column and demonstrated separation of bovine serum albumin (BSA) from lactoferrin. Similar to CAC [136], CRFC can be potentially integrated with a perfusion bioreactor; however, to date, commercial-scale CRFC systems do not appear to be available for bioprocessing applications.

2.2.2 Continuous Countercurrent Chromatography

Countercurrent processes inherently have superior mass transfer characteristics and thus high-purity products can be obtained, even at high flow rates. Likewise, countercurrent chromatography allows the stationary phase to be used more efficiently, improving productivity and decreasing solvent/buffer consumption as

compared to batch chromatography. The liquid-solid countercurrent movement is obtained by rotating the solid phase adsorbent (chromatography resin) in the direction opposite to the liquid phase flow. This, in principle, can be achieved by physically moving the solid adsorbent/columns in the opposite direction to the liquid flow (true moving bed) or by simulating the continuous movement of the solid phase using elaborate column and valve movements in a periodic or semi-continuous manner (simulated moving bed).

Simulated Moving Bed (SMB) Chromatography

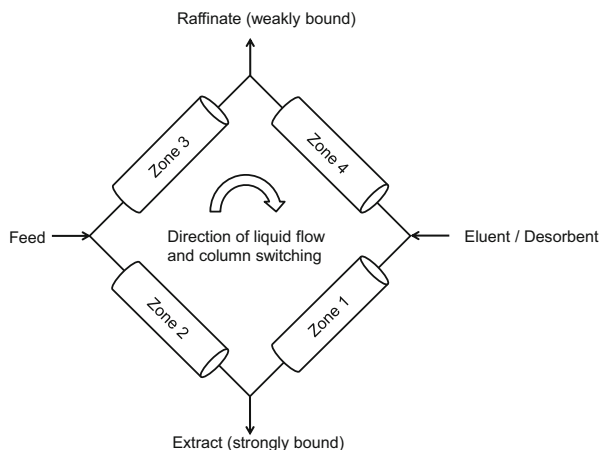
The SMB process principle is a flow scheme that takes advantage of continuous and countercurrent movement of liquid and solid phase without actual movement of the adsorbent. Instead of a large packed bed used in traditional batch chromatography, SMB chromatography systems are characterized by the connection of multiple smaller packed beds. The simulated countercurrent flow is carried out through a complex valving system located between the columns, such that inlet and outlet positions are periodically switched from column to column in the direction of liquid flow.

The petroleum industry originally developed the SMB in the late 1950s to separate *p*-xylene from its isomers [165]. Later, its application scope was extended to the sugar industry and by the 1990s it was used by the pharmaceutical industry for manufacturing chiral and biological drugs [166–168]. The technology development in all these process industries was motivated by the need for cost reduction and large-volume production of low-value chemicals at high purity.

In the classical four-zone SMB process, the solid phase is divided into four zones (Fig. 7). Zones 2 and 3 are separation zones and zones 1 and 4 are used to regenerate the adsorbent and the eluent. Only one solvent/eluent is used and hence this operation is in an isocratic mode. The feed and eluent streams are simultaneously applied and withdrawn at appropriate points between the columns, which causes the slow and fast moving components to travel in opposite directions relative to the feed port. The component with lower affinity toward solid phase (faster moving component) travels with the liquid phase, and purified substance is collected at the raffinate port. The component with higher affinity toward the solid phase (slow moving component) travels in the direction in which solid phase movement is simulated, and purified substance is collected at the extract port. Successful SMB design and operation depends upon the correct selection of operating conditions, zone liquid flow rates, and switching time (which corresponds to solid flow rates). The use of SMB operation in isocratic/classical mode is most relevant for SEC-based applications [6, 169].

The original design of the SMB system was suited for the separation of binary mixtures. Over the years, several variants of SMB have been introduced that allow separation of ternary mixtures and application of gradients [6, 170–173]. Simulated moving bed chromatography has been used to purify a variety of biological molecules, including mAbs from cell culture supernatant [174, 175], influenza

Fig. 7 Schematic of a four-column simulated moving bed (SMB) chromatography system



virus from cell culture supernatant [176], and lactoperoxidase and lactoferrin from whey protein concentrate [177]. A more detailed review of SMB chromatography and its applications have recently been discussed elsewhere [167, 178].

Periodic Countercurrent Chromatography

A simplified version of simulated moving bed chromatography is periodic countercurrent chromatography (PCC). This approach involves the use of three or more columns, where loading occurs in a countercurrent manner. A schematic of a cyclic three-column PCC system is shown in Fig. 8. The first column is loaded to saturation and the breakthrough is loaded onto the second column. The saturated column is washed on the third column and then eluted similar to a batch operation. Because such serial column loading loses very little of the target protein, even at high feed flow rates, column sizes can be considerably smaller than equivalent batch systems [17, 179]. The use of PCC is most suitable in capture applications where continuous feed stream is present (e.g., continuous clarified harvest from perfusion reactor), especially for labile molecules [17, 180]. To allow continuous operation, the duration of the recovery and regeneration tasks must not exceed the time for protein loading.

Bench-scale PCC units for biopharmaceutical application have been developed by GE Healthcare. These units employ a UV-based switching logic (where the key signal is the relative difference in UV at the column inlet and outlet) which allows monitoring and control of the column saturation level. This dynamic UV-based control makes the system highly responsive to variability in the feed stream or column performance. Successful integration of a perfusion bioreactor with PCC capture has been demonstrated using both stable (mAbs) and relatively less stable molecules (recombinant enzymes) [17, 180].

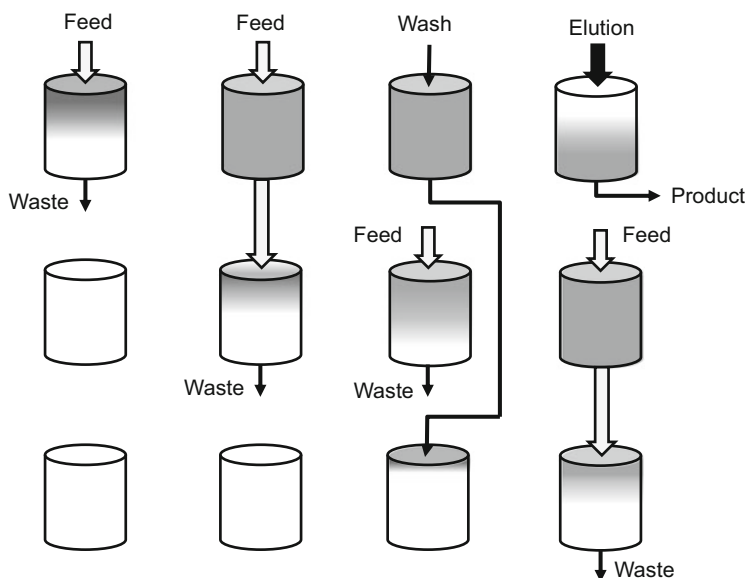


Fig. 8 Schematic diagram of the three-column PCC cycle

The use of CaptureSMB (ChromaCon), a two-column system analogous to PCC for mAb capture, has also been demonstrated [181, 182]. Instead of UV-based loading (used in GE PCC), the CaptureSMB system uses a time-based loading strategy to calculate breakthrough. To achieve sequential washing between two columns, the feed flow rate has to be zero; hence, a small reservoir bag between the continuous upstream processes and the CaptureSMB system is required. An advantage compared to other multi-column systems is that CaptureSMB requires less complex hardware, potentially leading to lower equipment costs and risks of failure. Other analogous multi-column chromatography systems currently available for biopharmaceutical applications are BioSC marketed by NovaSep [183], BioSMB by Tarpon Biosystems-Pall [184], and Octave by Semba Biosciences [185]. All these systems provide periodic (cyclical) operation, and hence are amenable to continuous operations.

Continuous Countercurrent Tangential Chromatography

Continuous countercurrent tangential chromatography (CCTC) is a recent development that allows truly continuous and countercurrent operation by enabling “true moving bed” technology, as opposed to simulated moving bed [186]. Instead of packing the resin particles in a fixed-bed column, the resin (in the form of a slurry) flows through a series of static mixers and hollow-fiber membrane modules. The microporous hollow-fiber membranes retain the large resin particles and let all dissolved species (protein and buffer components) pass through the membrane and

into the permeate. The buffers used in the binding, washing, elution, stripping, and equilibration steps flow countercurrent to the resin slurry in a multi-stage configuration.

In contrast to SMB, which operates in a cyclic mode, CCTC provides the opportunity to run at steady state without the need to switch valves after system start-up. As the resin is used in slurry form, the pressure drop is independent of particle size, in contrast to packed beds. Thus, smaller resin particles can be used in CCTC, allowing for higher mass transfer rates and improved productivity [187]. Another advantage is that all activities related to handling packed beds, including packing, cleaning, validation, and storage, could be eliminated. The concentration of product from a CCTC system is relatively dilute compared to packed bed systems. Hence, an inline concentration system may be needed to increase the final product pool concentration and decrease the loading volume for further downstream operations.

To date, proof of concept has been demonstrated for purification of BSA and myoglobin using an anion exchange resin [188], purification of an IgG4 from a mixture of BSA and myoglobin using a Protein A affinity resin [189], and purification of mAbs produced in Chinese hamster ovary (CHO) cells from clarified cell culture fluid [187, 190].

Multicolumn Countercurrent Solvent Gradient Purification

Multicolumn countercurrent solvent gradient purification (MCSGP) is a multi-column chromatographic process capable of using linear gradients for high-resolution separation of three component fractions [191, 192]. This strategy can be operated in a continuous [191] or semi-continuous [193] mode, with six columns or three columns, respectively. Similar to PCC and SMB, MCSGP uses multiple columns in sequence that are switched in position to get countercurrent contact. In contrast to traditional SMB, MCSGP can be used for more than just binary separations. For a complex mixture, where target product overlaps with early and late eluting impurities, three- or four-zone SMB cannot deliver pure target product in a single step; in fact, two sequential SMBs are needed to achieve this. Alternatively, MCSGP can be used for such difficult separations to obtain target product in a single step. Such separation challenges are typically faced in polishing steps of biopharmaceuticals. For example, Müller-Späth et al. [194] demonstrated the effectiveness of continuous MCSGP using cation exchange resins for the separation of charge variants of several commercial mAbs. Continuous capture via MCSGP has also been demonstrated. Müller-Späth et al. [195] demonstrated the use of a four-column MCSGP system with clean-in-place capabilities and fully continuous loading for the capture of mAb from clarified cell culture fluid using cation exchange resins with a gradient elution. Bench-scale MCSGP purification can be performed with a two-column system marketed as Contichrom (Knauer-ChromaCon AG).

2.2.3 Flow-Through Chromatography

In flow-through chromatography, impurities bind and the product of interest flows through. Anion exchange (AEX) products in flow-through mode have been widely used as a polishing step in mAb purification processes to remove residual impurities such as host cell protein, DNA, and a variety of viruses [196, 197]. Flow-through chromatography can be performed with conventional bead-based resin or membrane adsorbers [198, 199] and monoliths [200, 201]. In flow-through mode, membranes have shown advantages over traditional packed beds including faster operating flow rates, reduced buffer requirements, and disposability [202].

Use of flow-through operations in batch processes have improved productivity in feeds where impurities are less abundant than the product (polishing steps) or in processes with feeds at high product concentrations. The application of flow-through chromatography to continuous operations can be easily imagined with parallel or redundant units used in an alternating fashion; that is, one unit is binding impurities and the other column is being regenerated for the next cycle [16]. Such an approach can also be applied to other dead-end filtration systems.

2.2.4 Expanded-Bed Chromatography

Expanded-bed chromatography (EBC) utilizes a fluidized chromatographic adsorbent bed, which allows desired proteins to be purified directly from crude (particulate-containing) feedstock [203–206]. The primary difference in setup of EBC compared to traditional chromatography columns is that the top flow adapter is appropriately positioned such that the bed is allowed to expand upward in the direction of liquid flow. The bed expansion increases the bed voidage, which allows unhindered passage of cells, cell debris, and other particulates during application of crude feed to the column. Thus, a combination of clarification, concentration, and adsorptive purification can be achieved in a single step, allowing significant cost savings [158, 207, 208]. The flow distributor and the adsorbent/beads are the two main factors influencing the EBC operations [209, 210]. Batch EBC has been successfully used for protein purification directly from *Escherichia coli* homogenate, yeast, whole hybridoma fermentation, mammalian cell culture, milk, and animal tissue extracts [203, 206, 211–213].

Continuous countercurrent EBC has been used for direct purification of lysozyme from egg whites and enriched bovine milk [214]. The mode of operation consisted of four countercurrent contactors arranged in a series and in a loop for continuous loading, washing, elution, and regeneration. The process buffers and unclarified feed entered the respective columns at the base and moved upward through the adsorbent. The adsorbent bed was fluidized so that it behaved as an expanded bed instead of a well-mixed system. Continuous operation was obtained by constantly removing adsorbent through the base of the bed and simultaneously applying it to the top of the next stage, just above the level of the expanded bed. In

conventional batch EBC, each adsorbent particle remains approximately stationary at a point governed by parameters such as liquid flow rate and particle dimensions. Conversely, in the continuous countercurrent EBC system, removal of adsorbent from the base of the bed causes suspended particles to fall with respect to the column, thus resulting in a net downward flow of the adsorbent material [215]. Similar to CCTC, a truly countercurrent continuous operation can be achieved with continuous product streams generated from unclarified feeds. Batch EBC columns have recently been made available (DSM RhobustTechnology) but application of continuous EBC for routine biological production has not been reported.

2.3 Other Technologies for Continuous Purification

2.3.1 Continuous Clarification of Cell Culture Harvest

Many of the perfusion-enabling cell retention methods previously mentioned can also be used for stand-alone continuous clarification (e.g., gravitational settling, centrifugation, acoustic settling, and tangential flow filtration). Continuous depth filtration has also been demonstrated in a manufacturing environment using redundant filter trains. Instead of directly processing cell-containing bioreactor material, the downstream system was designed to process continuously harvest material that had passed through a gravitational settler. Parallel filter trains were integrated into an automated control system that switched to a new train when the current train reached a high pressure limit [16, 52].

2.3.2 Continuous Precipitation and Flocculation

Precipitation and flocculation have been explored to improve purification efficiency and impurity clearance and to simplify the process by decreasing the number of purification steps [196, 216, 217]. Compared to chromatography, these techniques provide greater flexibility across a range of protein titers as scale-up is volume-based and independent of titer. Precipitation methods rely on lowering the solubility of target molecules to create solid particles. This can be achieved by altering factors such as pH and conductivity (salting-in or salting-out) of solution, or through addition of precipitants such as ethanol, ammonium sulfate, polyethylene glycol (PEG), caprylic acid, or divalent ions [218–222]. Flocculation relies on flocculating agents such as polyelectrolytes that cause adhesion of dispersed particulates into larger-sized clusters, resulting in an increase in the average particle size [223, 224]. The flocs formed can be collected by filtration, sedimentation, or centrifugation at low speed. Precipitation/flocculation methods are designed either to precipitate or flocculate the protein of interest, leaving impurities (DNA, host cell proteins, cell debris, etc.) in solution or to precipitate impurities and leaving the protein product in solution [205, 225].

Precipitation has been used at industrial scale for blood plasma protein fractionation. However, this application has been typically performed in batch mode in several stages where ethanol is added to aqueous solutions at specified conditions for pH, temperature, and ionic strength [226]. Continuous fractionation of human blood plasma using tubular reactors [227, 228] and a series of mixed-suspension mixed-product removal reactors (MSMPR) [229] has been demonstrated. Along with tubular reactors and MSMPR, continuous precipitation of proteins can also be performed using centrifugal precipitation chromatography [219].

Hammerschmidt et al. [230] converted a two-stage batch precipitation system based on calcium chloride flocculation and cold ethanol precipitation into a continuous purification process for recombinant antibodies from clarified cell culture supernatant. Tubular reactors with cooling (via a double pipe heat exchanger) and helical static mixers were used for continuous operation. The continuous system not only provided uniform precipitate yields and improved productivity but also offered uniform cooling, which can become a limiting factor during cold ethanol batch precipitation processes. Similarly, continuous sequential precipitation of impurities followed by a target monoclonal antibody (mAb) in clarified cell culture supernatant with PEG as precipitant has also been demonstrated [231]. Inexpensive precipitants (ethanol, calcium chloride, PEG, etc.) can be cost-effective replacements for the Protein A capture step, providing an opportunity to develop inexpensive fully continuous mAb processes [10, 232].

2.3.3 Continuous Aqueous Two-Phase Extraction (ATPE)

Aqueous two-phase extractions (ATPE) are liquid-liquid biphasic systems that have been used for downstream processing of several biological products including cells, proteins, enzymes, hormones, viruses, and plasmid DNA [233–238]. Aqueous two-phase extraction can be applied for clarification purposes [239] and can also serve as an alternative to more traditional chromatography operations [240]. ATPE relies on the formation of two immiscible aqueous phases such that the target protein partitions preferably into one phase and impurities partition preferably into the other. Organic solvent mixtures are not suitable as they can induce protein denaturation. Hence, ATPE for biological compounds is usually carried out with two polymers or a polymer and a salt dissolved in water above a critical concentration [240]. Polymers (such as PEG, polyvinyl alcohol, dextran, and starch) and buffering salts (such as phosphates, sulfates, and citrates) have been used to create the two phases. To improve the specificity of the protein of interest to a particular phase, functionalization of the phase-forming component is also possible [241].

The mechanism of phase separation in aqueous two-phase systems and in conventional organic–aqueous systems is similar; hence, most of the standard extraction equipment used in the chemical industry can be used for ATPE [242, 243]. Continuous ATPE can be performed using appropriate column contactors, spray columns, rotating disk contactors, or mixer-settler units [244–246]. Use of multi-stage ATPE devices such as multi-plate column contactors and

mixer-settler configurations can overcome the limitations of single-stage extraction through improved recovery and/or purity [247, 248]. Multi-stage ATPE has been successfully evaluated for the purification of mAbs [245, 248–250]. Continuous ATPE devices provide a promising non-chromatographic alternative for use in downstream mAb processing, allowing simultaneously clarification, concentration, and partial purification in one unit. One of the concerns about ATPE at industrial scale is the high consumption of polymers and salt leading to negative impacts on water treatment [247].

2.4 Continuous Formulation

Formulation of therapeutic proteins is developed based on clinical needs, patient compliance, delivery method, stability of the drug, storage and distribution, and market competitiveness. Liquid formulations have generally been preferred as they are less expensive, faster to develop, and generally more convenient for administration. However, dried formulations (e.g., lyophilized or crystallized) have been used successfully for protein drugs that may not be stable enough to be handled as a liquid formulation.

2.4.1 Continuous Ultrafiltration and Diafiltration

Ultrafiltration/diafiltration (UF/DF) is routinely used in the biopharmaceutical industry for high concentration liquid formulation of therapeutic proteins. Highly concentrated protein formulations are desirable for therapeutic uses as they allow for dosages with smaller volumes, and are more economically packaged and stored. The use of a UF/DF system allows both concentration of the product and diafiltration into desired buffer (e.g., formulation buffer) to be performed with the same unit. Typically, UF/DF uses tangential flow filtration technology, where feed flows parallel to the membrane surface rather than perpendicular to the surface. Traditionally these steps are operated in batch mode. During ultrafiltration, the feed is recirculated through the membrane module and multiple passes are required to achieve the desired concentration. During diafiltration, the feed is recirculated through the membrane module as the formulation buffer is added at a rate equal to the permeate flow rate to maintain constant volume. Approaches to continuous UF and continuous DF operations have been described in the literature.

Single-pass tangential flow filtration (SPTFF) concentrates protein solutions in a single pass, without the need for a recirculation loop and tank [251]. The feed is directly pumped through the SPTFF module and concentrated product is obtained from the retentate port. The basic principle of SPTFF is increased residence time in the feed channel, which leads to increased conversion in a single pass through the module. Increased residence time can be achieved by reducing the flow rate, increasing the path length in a serial configuration or by simply adding more TFF

membranes in series. Single-pass tangential flow filtration is designed to have multiple conventional TFF stages in a single unit, which can potentially minimize capital investment and footprint area compared to traditional TFF systems. The use of SPTFF can easily be implemented in integrated and continuous processes as inline concentrator for dilute streams [252–254] or in the final formulation step for concentrated liquid formulation.

Continuous DF using TFF can be achieved with a multi-stage cascade system where DF occurs in either a concurrent or countercurrent manner [6, 255]. Multi-stage cascade UF systems have been successfully used for continuous protein purification [256–258]. Similar to cascade UF systems, for multi-stage concurrent DF processes multiple filtration units are connected in series. In the first stage, feed and buffer are added at a rate equal to the permeate flux. In the subsequent stages, only fresh buffer is added at a rate equal to permeate flux. The countercurrent DF process also consists of several units in series, but enables lower buffer utilization. Here, feed is continuously added to the final stage and fresh buffer is continuously added to the first stage. The retentate from each stage is recycled and serves as a feed to the previous stage in the cascade. As the numbers of stages increase, the impurity removal improves and the buffer requirement decreases [9]. One of the challenges to using membrane cascades is the delicate control of interacting flows [259, 260]. In addition, with an increasing number of membrane stages, extra buffer tanks and pumps are usually required, complicating the process and increasing the initial capital cost. Therefore, for a membrane cascade system to be practical, the number of cascade units must be minimized and the control of the system needs to be simple.

Other approaches to industrial-scale buffer exchange systems are through use of SEC and countercurrent dialysis [261]. Size-exclusion chromatography for continuous buffer exchange can be envisioned using a simulated moving bed system with multiple columns. Klutz et al. [16] used a dialysis membrane module, where continuous buffer exchange was performed in a countercurrent flow. The feed stream was continuously fed through the hollow fibers/capillaries, whereas the wash fluid/diafiltration buffer was passed through the shell side. The flow rate of the feed and retentate stream (fed through the capillary) and of the diafiltration buffer was controlled through use of pumps. Gambro Revaclear 300 capillary dialyzer (Baxter) was used for performing continuous diafiltration; however, use of other hollow-fiber devices is also possible [262].

2.4.2 Continuous Lyophilization

Lyophilization (i.e., freeze-drying) is the method of choice to increase the shelf life of sensitive proteins. By removing water from the formulation and sealing the resulting cake in a vial, the drug can be stored, shipped, and later reconstituted to its original form for injection. Lyophilization is inherently conducted as a batch operation. The starting liquid solutions are filled in vials or trays, frozen, dried (the remaining “bound” water is driven out), and then handled as a batch.

To date, continuous lyophilization for biological products has not been demonstrated. The food industry, which routinely uses freeze-drying, has taken steps to move from batch to semi-continuous or continuous freeze-drying approaches because of the large production needs and cost-effectiveness. GEA Niro has developed a continuous lyophilization system (CONRAD) for food products (such as coffee, fruit, and meat) in which the product is moved through a long tubular lyophilization chamber on a tray.

Efforts to develop continuous freeze-drying for pharmaceutical industry are ongoing. De Meyer et al. [263] evaluated spin freezing instead of conventional freezing as a part of a continuous freeze-drying step. In spin-freezing, the vials with liquid formulation are rotated rapidly along their longitudinal axis. The cooling and freezing of the solution are achieved by using a flow of sterile gas with a controllable temperature around the rotating vial. Such frozen vials could be continuously transferred into separate drying units (primary and secondary dryer units) through appropriate load-locks which maintain pressure and temperature within each unit [263]. Weisselberg [264] has described a continuous lyophilization process in which the product solution cascades through a series of vertically stacked trays. Rey [265] has discussed the possibility of using similar systems for lyophilization of biological products, although this has not yet been demonstrated in practice.

2.4.3 Continuous Crystallization

Protein crystallization, which has been mostly applied in protein structure analysis, has also been recognized as a method of protein purification [266, 267] and formulation [268]. Protein crystals form a regular lattice that can exclude impurities and misfolded protein, and hence can be used in purification applications. Feasibility of protein purification by crystallization has been demonstrated with lipase [269], ovalbumin [270], mAbs [271], and at industrial scale for insulin [272].

For formulation applications, the crystalline state of proteins offers several advantages for delivery: (1) longer shelf life and more stability than liquid formulations, (2) controlled release because of better dissolution characteristics, and (3) subcutaneous or inhalation therapy-based delivery instead of intravenous infusion [271, 273]. Protein crystallization for delivery and formulation application has been extensively reviewed [268, 274].

Crystallization is often operated in batch mode in chemical and pharmaceutical applications [275]. The most dominant crystallizer type used in pharmaceutical industry for batch operations is based on a stirred tank design [276]. Continuous crystallization offers an opportunity to provide enhanced product quality, improved process efficiency, and better control of crystal morphology [277, 278]. In recent years, continuous crystallization has attracted great interest in the pharmaceutical industry [279–281]. As shown in Fig. 9, there are two main types of continuous crystallizers that have been investigated: (1) mixed-suspension mixed-product removal (MSMPR) reactors and (2) tubular crystallizers such as plug flow reactors (PFR) and continuous oscillatory baffled crystallizers (COBC) [276]. MSMPR

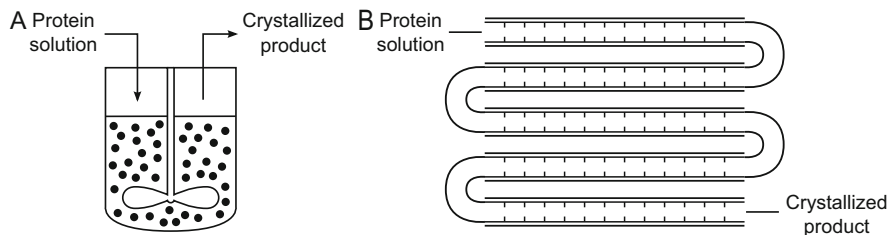


Fig. 9 Schematic description of (A) a mixed-suspension mixed-product removal (MSMPR) reactor, and (B) a continuous oscillatory baffled crystallizer (COBC), which serves as a specific example of a tubular flow reactor [277]

reactors have been widely studied in the pharmaceutical industry. This system operates similar to continuous stirred tank reactors where product is continuously introduced and crystals are continuously removed. Converting batch operations to continuous mode by operating the stirred tank designs in MSMPR mode is considered to be an easier and more cost-effective approach than replacing existing batch equipment with completely new continuous crystallizer designs [276]. Several configurations of MSMPR reactors have been developed (single-stage, multi-stage, with and without recycling operations, etc.).

Plug flow reactors rely on tubular reactors with static mixers, and offer an alternative means of continuous crystallization. Such crystallizers can provide continuous crystallization with much more uniform product quality (i.e., narrow crystal size distribution) [275]. Plug flow crystallizers are ideally suited for mechanically fragile systems such as proteins over tank-based crystallizers; however, long tubes may be required to attain sufficient residence times. A COBC is a tubular crystallizer containing periodically spaced orifice baffles, which allows uniform mixing because of the creation of oscillatory motions in each interbaffle zone [277]. Scale-up is potentially easier in tubular crystallizers as high product quantities can be obtained by installation of parallel tubes and/or running the systems for longer times. Recently, Neugebauer and Khinast demonstrated a continuously operated tubular crystallizer for the production of lysozyme protein crystals [282]. However, commercial application of continuous crystallization for recombinant protein production has yet to be established.

2.5 Continuous Viral Inactivation and Removal

To ensure viral safety, regulatory guidelines require that manufacturers of biological products for human use demonstrate the capability of the manufacturing process to remove and/or inactivate known or adventitious viruses. Typically, removal is established through virus reduction filters and chromatography steps, and inactivation is established by use of heat, irradiation, chemical, or pH treatment.

2.5.1 Virus Removal by Filtration

Virus reduction filters (i.e., nanofilters) remove viruses from the product stream using size exclusion [283, 284]. The main advantage of this process is that, under optimal conditions, both enveloped and non-enveloped viruses can be robustly removed without affecting the quality of the derived products. Traditionally, viral filtration has been operated in batch mode where each batch of feed material is processed sequentially through the viral filter. To date, viral filtration technology has not been able to process a continuous input of feed stream for prolonged periods of time. However, similar to flow-through membranes and flow-through chromatography, continuous viral filtration is possible using redundant and parallel filters. The feed can be switched to the second filter when the capacity of the first filter is achieved [285].

2.5.2 Continuous Viral Inactivation

In a traditional monoclonal antibody process, inactivation of enveloped viruses is achieved by incubation at low pH [286, 287]. This is inherently a batch process, where a process intermediate is collected in a vessel and held at low pH for the required inactivation time of 30–120 min [286, 288]. To date, commercial application of continuous low pH viral inactivation has not been reported. Several configurations of tubular reactors have been proposed for continuous viral inactivation, and Klutz et al. [285, 289] have demonstrated the most promising among these to be the coiled flow inverter (CFI). A CFI reactor consists of several straight helix modules, where the tube reactor is coiled around a coil tube. This reactor provides close-to-plug-flow behavior in a laminar flow regime and long residence times. The successful proof-of-concept was demonstrated with Protein A capture eluate continuously processed through a CFI [285]. CFI reactors have compact design, narrow residence time distribution, low investment costs, easy scale-up, wide operating window, and compatibility with single-use technology [285, 289]. Similarly, use of CFI could be adopted for continuously operated inactivation using chemical treatment (e.g., solvent/detergent).

Ultraviolet-C (UV-C) light at a wavelength of 254 nm is an effective method for inactivation of non-enveloped viruses [290, 291]. Continuous flow UV-C reactors for virus inactivation have been designed [292, 293]. The effectiveness of continuous flow UV-C in viral inactivation of protein solutions has also been demonstrated [294–296]. A continuous UV-C system, UVivatec, was developed by Bayer Technology Services and is marketed by Sartorius Stedim Biotech. Similar to continuously operated inactivation, in continuous UV-C, fluid streams move helically along the lamp. Secondary circulating flows (Dean vortices) are generated which provide highly efficient mixing and allow for uniform and controllable irradiation of the entire volume.

3 Case Studies for Integrated Continuous Biomanufacturing

Several approaches to integrated continuous biomanufacturing are possible, including hybrid systems (i.e., some unit operations are integrated and continuous whereas others are conventional batch operations) and fully integrated continuous systems [7]. In most cases, integration and continuous operation provide increased productivity and flexibility, decreased cost of operations, and improved product quality (especially for labile molecules). Fully continuous systems potentially offer the maximum benefits among these approaches. To date, commercial implementation of fully integrated continuous processes has not been established. However, several proof-of-concept approaches have been attempted which provide opportunities for the implementation of end-to-end continuous processes in the biopharmaceutical industry.

Vogel et al. [52] demonstrated a hybrid integrated manufacturing platform, where an upstream perfusion bioreactor was integrated with a continuous cell separation process based on disposable flow paths, followed by membrane chromatography operated in a rapid cycling mode. Further purification/processing was performed in a batch mode. The continuous perfusion reactor was equipped with an inclined plate settler for cell retention, returning a majority of the cells back to the reactor. Harvest (along with some debris) was further processed through continuous cell separation via two parallel disposable filter assemblies operated in alternating fashion. Rapid capture of the target proteins (recombinant blood coagulation factors) in clarified harvest was achieved through large scale anion exchange membrane adsorbers. Overall, purification performance with the integrated platform process was comparable to the conventional process, with a significant reduction in downstream cycle time. A similar integrated approach (upstream perfusion, continuous cell separation, and membrane chromatography capture) has been successfully used in GMP manufacturing for other molecules at Bayer.

As depicted in Fig. 10, lab-scale end-to-end fully continuous production of mAbs has been demonstrated by Godawat et al. [15]. This process included a perfusion bioreactor with a cell retention system (ATF) and two periodic counter-current chromatography systems (PCC1 and PCC2). This work built upon a previously demonstrated integrated continuous capture system consisting of a bioreactor and PCC [17, 180]. In the end-to-end approach, the bioreactor/PCC1/PCC2 system was run in an automated and continuous manner across the entire upstream and downstream operations, ultimately producing drug substance. The continuous operation was enabled by balancing the rate of mass and flow across various unit operations. PCC1 was used for continuous capture using Protein A resin. The low pH inactivation was performed in a cycled batch mode. The intermediate column eluate was designed such that the protein eluted in appropriate formulation buffer at the required drug substance concentration. The eluate was further polished using a membrane adsorber in a flow-through mode, resulting in minimal changes to either the protein concentration or buffer formulation, thus eliminating the requirement

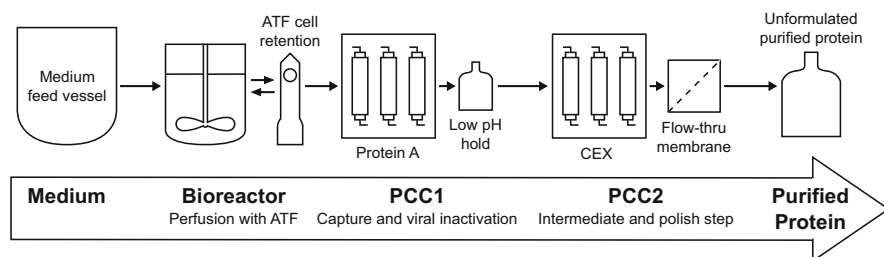


Fig. 10 Architecture for one example of end-to-end continuous bioprocessing from Godawat et al. [15]

for terminal UF/DF. However, this approach might not be feasible for all protein drugs, and incorporation of UF/DF in the downstream architecture may be required. This end-to-end bioprocessing architecture did not include a nanofilter in the process train because the pumps and the valves available on the two PCC systems did not allow for addition of another unit operation. The data in the article indicated that the end-to-end process increases process throughput and significantly decreases the equipment footprint, eliminating several non-value added unit operations.

Another proof-of-concept for fully continuous and automated production of mAbs has been demonstrated by Klutz et al. [16] (Fig. 11). This process included a perfusion bioreactor with continuously operated filtration achieved by redundant filters. Protein A capture chromatography was performed using a multi-column chromatography system (BioSMB). A continuous flow reactor with close-to-plug-flow behavior, the CFI reactor, was used to provide the necessary incubation time of 60 min for viral inactivation at low pH. Continuously operated intermediate purification and polishing was performed with two subsequent chromatography steps in flow-through mode. Viral filtration was operated with two alternating filters. The formulation was achieved with a continuous concentration and diafiltration. An ultrafiltration membrane was used for concentration, whereas countercurrent diafiltration using dialysis membrane modules was used for buffer exchange. Disposable bags were used as a reservoir between every unit operation to increase process robustness and to balance short-term differences in process flow rates between successive unit operations. The upstream unit operations prior to multi-column Protein A chromatography were operated continuously for 28 days; however, downstream unit operations were only operated for 60 h (because of sterility concerns). Though robust long-term operation was not achieved, this study demonstrated a successful technical proof-of-concept for a fully continuous monoclonal antibody process.

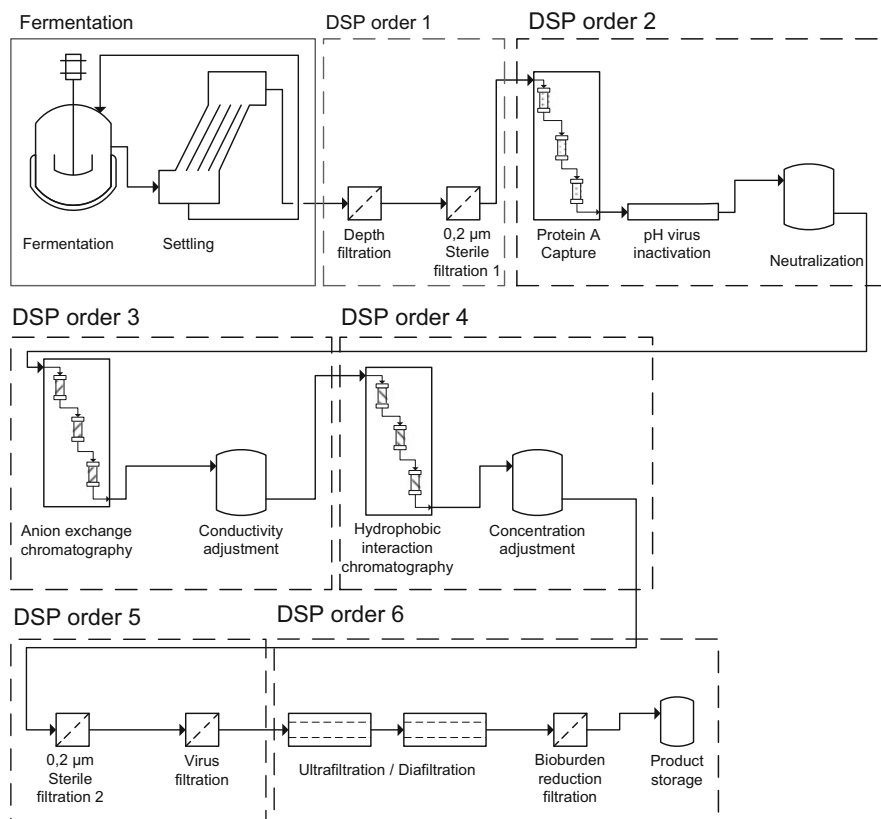


Fig. 11 Process scheme for continuously operated production of monoclonal antibodies as described by Klutz et al. [16]. Picture adapted from Klutz et al. [11]

4 Conclusion and Outlook

Continuous manufacturing has been well-developed in the petrochemical, chemical, and food industries. Recently a growing number of pharmaceutical companies have been working toward building continuous manufacturing into their processes. The biopharmaceutical industry is gradually maturing and continuous manufacturing of recombinant therapeutic proteins is still in its infancy. Economic pressures from increased competition and government regulation, coupled with the desire to gain access to emerging markets, are driving process innovation and implementation of continuous bioprocessing in this industry. As discussed in this review and by others in the literature, most of the necessary building blocks to implement a fully integrated and continuous bioprocessing platform are already established. The case studies highlighted in this review are significant advances toward realization of fully continuous integrated processes for monoclonal antibodies. However, there are still technical challenges that must be overcome before successful commercial

implementation. These challenges include better online process analytical technologies, cell line stability, reliability of hardware for long term operability and sterility, reliability of scale-down models, robust automation, and global process control for all unit operations. These technical challenges can be addressed with joint effort and collaboration between industry, academia and, regulatory agencies. Although it may not be straightforward to transition from batch to continuous manufacturing, there are significant long term benefits that could prove crucial as the biopharmaceutical industry evolves.

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